Status: Research Note / Empirical Validation
Epistemic Level: Conceptually complete, Evidentially pending (Reproducibility Fixture Needed)
Core Result: Bifurcation of float failure into Metric Lock (ULP) and Syntactic Saturation (Apeirogon). 1-ULP tie verified.

Metric vs. Syntactic Saturation: A Fare Algebra Audit of Heron's Method

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Instrument: fare-audit (RTFA v1.0)
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1. Introduction

When an iterative algorithm approaches an irrational limit, precision loss is traditionally treated as a local, gradual degradation.

The MBP framework (look for sources at the end) approaches this differently: numeric representations are modeled as "moveset environments" (algebraic systems with strictly defined sets of allowed operations). When an operation exceeds an environment's capacity, a "Fare" is paid and recorded in a tamper-evident exact ledger.

A critical question arises when a finite environment fails: is this failure uniform? This paper demonstrates that float saturation bifurcates into two distinct mechanisms:

1. **Syntactic Saturation:** The operation itself is forbidden because it constructs an irrational value not supported by the environment's algebraic moveset. When this is forced, the environment hits an **Apeirogon limit** - a geometric boundary where the finite grid is forced to approximate an infinite-sided continuous curve and fails.
2. **Metric Saturation:** The operation is legal (e.g., rational division), but the geometric grid of the target format (binary64) is too coarse to hold the exact result.

By applying the fare-audit instrument to Heron's method, we measure both phenomena precisely.

2. The Instrument and the Two Limits

The fare-audit engine computes exact rational results, forces a single correctly rounded cast into binary64, and records the exact Unit in the Last Place (ULP) distance. Crucially, the engine tracks the two distinct failure mechanisms:

- **The Drift Counter (Syntactic Saturation):** Fires only when an operation constructs an irrational exact value inside the float environment (e.g., sqrt or pow). At three nested irrational constructions, it halts, declaring an Apeirogon limit reached.
- **The ULP Bound (Metric Saturation):** Measures the exact distance from the exact rational to the nearest float. If an exact trajectory falls permanently into the rounding interval of a single float, it triggers a Representational Lock.

3. The Heron Experiment (Metric Saturation)

We apply the exact audit to Heron's method for the square root of 2: $x_{n+1} = (1/2) * (x_n + (2 / x_n))$

Starting from $x_0 = 1$, we track two trajectories: the single-cast (the exact rational cast once) and the float-pipeline (executed entirely in binary64). The exact numerical data presented in this section - including the exact rational limits, the detection of the 1-ULP tie, and the fractional convergences - were generated via a closed computational run of the Z.ai GLM engine, acting as an independent auditor against the fare-audit source code.

The Structural Cliff

The data reveals three regimes:

- **Step 1:** Exact (E). The iterate 3/2 is dyadic.
- **Steps 2-4 (Hunting):** The exact iterates require infinite binary expansions. The ULP distance fluctuates (+1/3, +19/51, -5407/29427) as the environment searches for a stable representation.
- **Steps 5-10 (Lock):** The float trajectory permanently freezes. The exact rational fraction, meanwhile, continues to expand exponentially (reaching 392 digits by step 10). The environment has hit the limit of its grid.

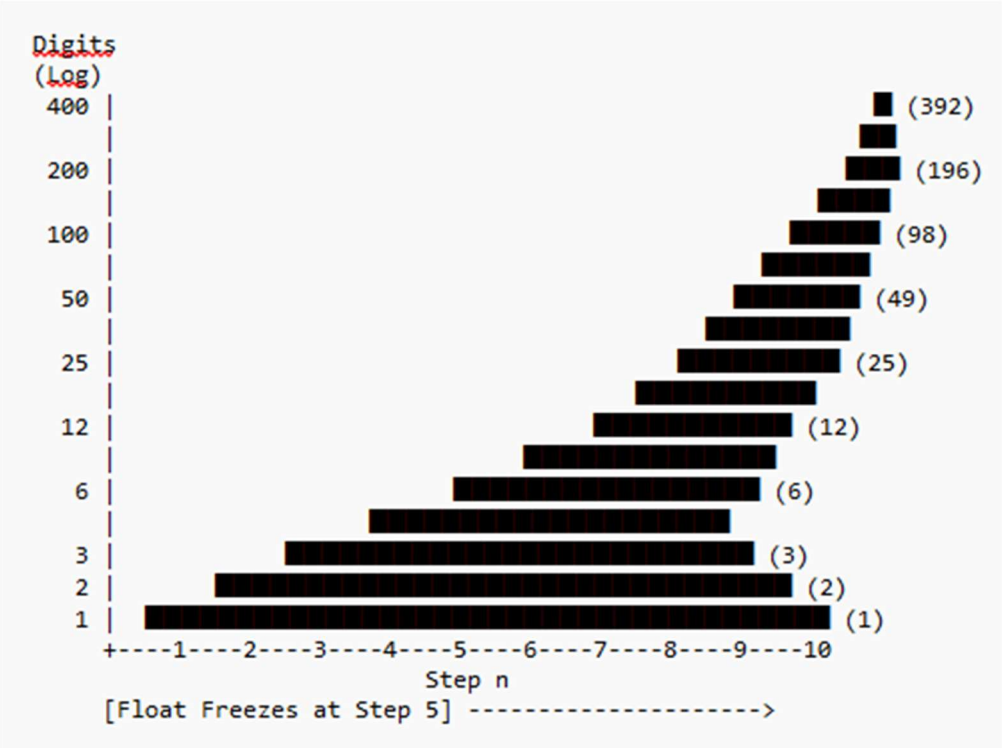


Figure 1. Decimal digits of the exact iterate (logarithmic scale) against the step number. While the binary64 environment locks its representation at 53 bits (Step 5 onward), the exact rational fraction continues to double in size at every step, reaching 392 digits by Step 10. This exponential memory growth is the exact "fare" paid to maintain rational certainty.

The 1-ULP Tie

Exact auditing reveals a microscopic divergence. The single cast freezes at 1.4142135623730951 (0x...bcd), paying 0.435376 ULP. The float pipeline freezes at 1.4142135623730949 (0x...bcc), paying 0.564624 ULP.

The two doubles are exactly one ULP apart. Because the float pipeline executes standard binary64 division and addition at every step, it accumulates intermediate rounding errors. The difference (0.129248 ULP) is the exact metric tax paid for intermediate rounding. The pipeline is trapped in a round-to-nearest-even tie at step 5, forcing it to resolve downward to the even mantissa.

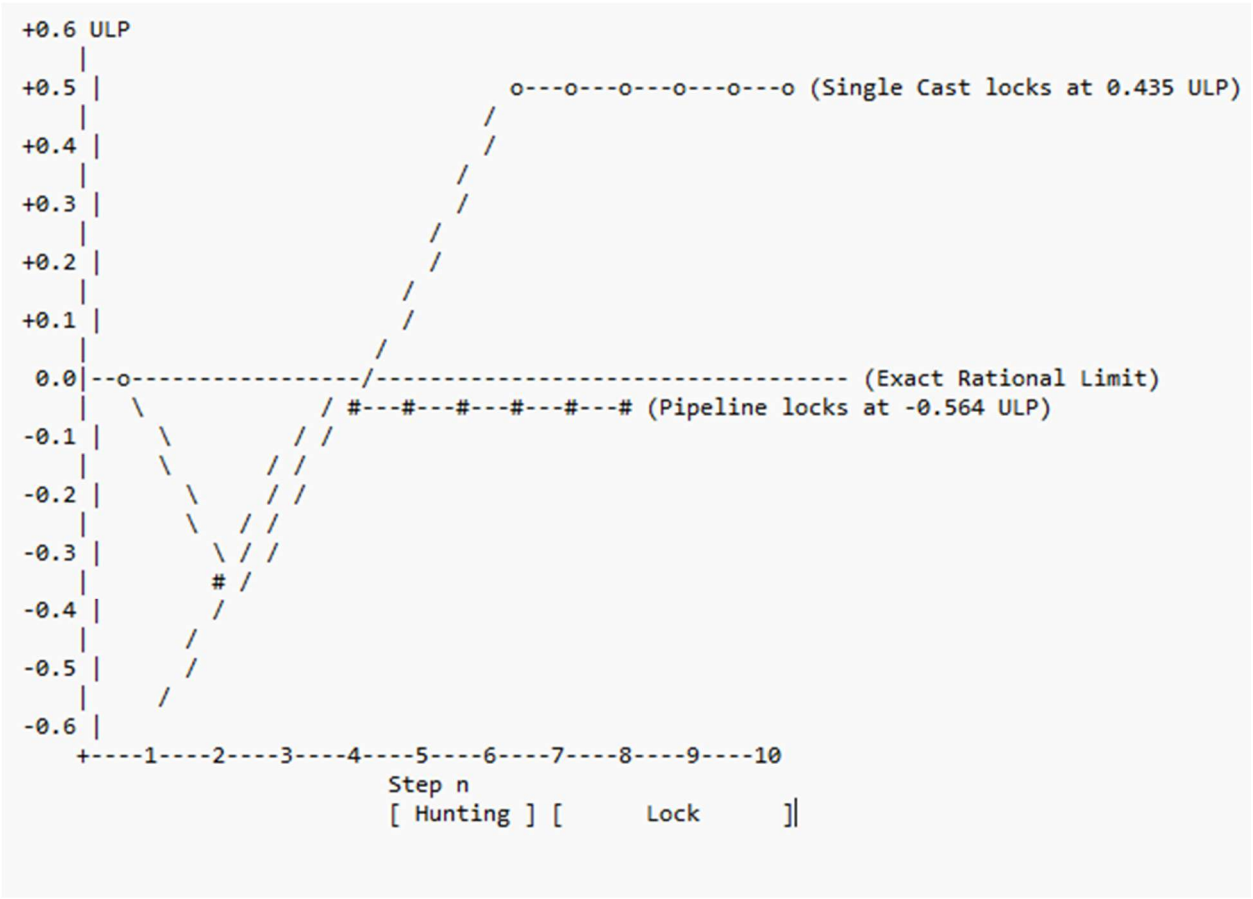


Figure 2. Signed distance from the exact iterate to each binary64 trajectory, measured in ULP. Both trajectories hunt for a stable representation (Steps 2-4) before permanently freezing at Step 5 - landing on neighboring doubles separated by exactly one ULP. The top trajectory represents the single-cast exact rational; the bottom trajectory represents the float pipeline.

The Exact Error Still Converges

The exact fractions (E5–E10) are six distinct rational numbers. They are strictly converging toward the limit |sqrt(2) - fl(sqrt(2))|, but the changes happen beyond the 33rd significant digit. What appears as a flatline at 9.667e-17 is merely a rendering artifact of three-significant-digit display.

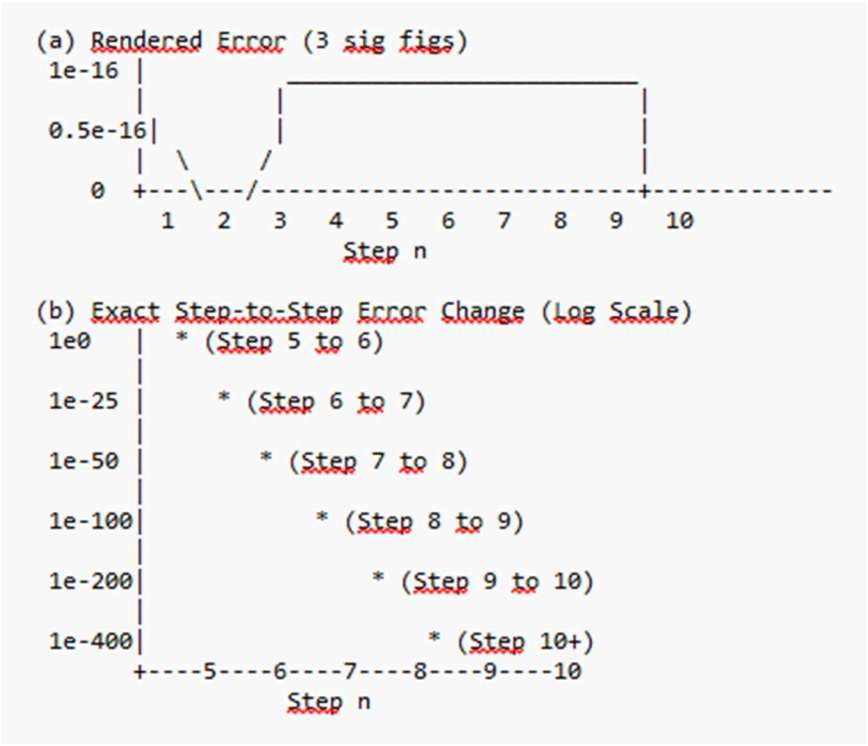


Figure 3. The flatline illusion. (a) The rendered absolute error appears completely flat at 9.667e-17 from Step 5 onward. (b) However, the exact step-to-step change of the same quantity reveals that the error is still shrinking quadratically, dropping to 3.085e-392 by Step 10. The convergence continues beneath the threshold of the three-significant-digit display.

4. The Apeirogon Experiment (Syntactic Saturation)

To prove that Syntactic Saturation is a distinct mechanism from the Metric Lock, we audit a contrast chain: taking the square root of 2 three times in succession.

- Step 1: F64:2 --sqrt--> F64. Drift: 1.
- Step 2: result --sqrt--> F64. Drift: 2.
- Step 3: result --sqrt--> F64. Drift: 3. **SATURATION EVENT (Apeirogon limit).**

Here, the environment halts not because of a metric tie, but because it lacks the moveset to construct the nested irrationals. The finite grid has shattered against the geometric requirement of the continuous curve. Heron's method, using only rational arithmetic, never triggers this counter.

5. The Bifurcation: Syntactic vs. Metric

This experiment forces a necessary refinement of the framework. The failure of a finite environment is not a single phenomenon. It is bifurcated:

1. **Syntactic Saturation (The Apeirogon limit):** The environment fails because the *operation itself* is forbidden by the moveset. It is a category error that hits the geometric limit of the continuous curve. The drift counter fires.
2. **Metric Saturation (The Representational Lock):** The operation is legal, but the *geometric grid* of the float is too coarse. The exact trajectory falls into a rounding interval and cannot escape. The ULP bound is hit.

   This distinction provides the exact vocabulary to separate a forbidden action from a geometrically trapped one.

6. The Bleed and the Black Box: A Research Question

The metric lock observed in F64 is the empirical measurement of what the MBP framework calls the *transfinite bleed* between discrete grids and continuous limits. The lack of a discrete numerical boundary for the square root of 2 is not an abstraction; it is the observable fact that the float environment is forced to arbitrarily choose a side (the 1-ULP tie) based on a hardware convention.

This raises a fundamental question regarding computational paradigms that operate beyond the 1D discrete real line. Quantum algorithms, such as Shor's, operate on the continuous phase of qubits (the complex plane). They do not resolve irrational limits by converging on a discrete grid; they map problems into continuous periodic structures where phase information is fully accessible.

We do not claim that the 0.129248 ULP tie-breaking tax is the exact obstacle that Shor's algorithm bypasses. However, we propose a structural resonance for future investigation: **If a real-only discrete environment inevitably locks due to metric saturation, what could we achieve if we treated the complex phase not just as a quantum state, but as a continuous algebraic oracle for resolving metric ties that discrete real-grid architectures cannot escape?**

7. Conclusion

The exact audit of Heron's method substantiates the structural reading of floating-point error and sharpens it. The cliff is real, but the exact ledger reveals two cliffs: the single cast and the float pipeline freeze at two different doubles, separated by a round-to-even tie. The observed wall is a Metric Lock, distinct from the geometric Apeirogon limit reached during Syntactic Saturation (which fires readily on nested irrationals). By separating the syntactic limits from the metric limits, the framework gains a precise vocabulary for mapping the exact boundaries of finite computation.

FAQ

Isn't this just standard IEEE 754 float precision loss?
The loss of precision itself is standard. What is novel is the instrumentation and the bifurcation. Classical analysis treats float drift as a local mathematical reality. This framework identifies two distinct structural topologies: Syntactic Saturation (hitting the geometric Apeirogon limit) and Metric Saturation (hitting the grid limit).

Why does the exact fraction size grow so fast?
Because Heron's method squares the denominator at each step as it converges on an irrational number. Maintaining exactness requires exponential memory, while the float environment locks at 64 bits.

Can I reproduce this?
Yes. The fare-audit code is open-source (MIT License) and fully automated. While the exact fractions and 1-ULP tie data in this paper were derived via a closed Z.ai GLM audit run, the engine and test suites used to generate the broader experiment are publicly available.

- Run the manual test: https://github.com/Bitriada/poetica-mathematica/actions
- View the source: https://github.com/Bitriada/poetica-mathematica/tree/main/5_Code/Fare_Audit
- Permanent DOI: https://doi.org/10.5281/zenodo.24088599
- MBP framework: https://github.com/Bitriada/poetica-mathematica/blob/main/3_Standalone_papers/Monkey_Business/Monkey_business_arithmetic_spectator_pipeline.pdf

Provenance
The theoretical framework, specifications, and experimental design are the work of the author. The initial fare-audit engine was implemented with the assistance of Z.ai GLM against the author's specification. The specific numerical results, exact rational fractions, and the discovery of the 1-ULP round-to-even tie presented in this manuscript were generated during a closed computational run by Z.ai GLM, acting as an independent auditor against the engine's source code. The public CI workflows (GitHub Actions) used to reproduce the broader test suite were generated with the assistance of DeepSeek.